MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: LOW MATING FORCE, FEMALE BRUSH, STRAIGHT-THROUGH, POLARIZED, (.100 SPACING)

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

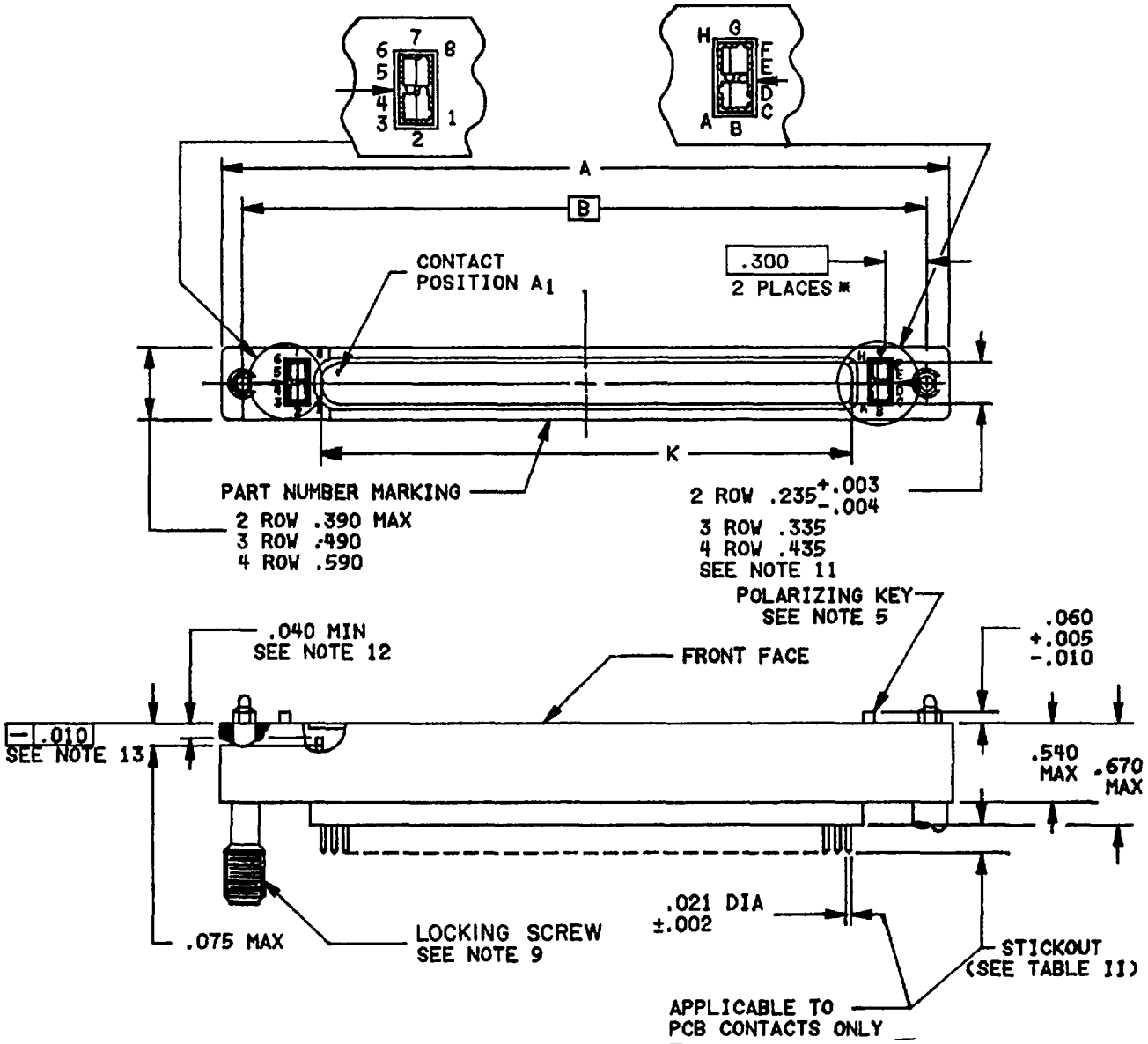


FIGURE 1. Connector, female brush, straight-through for multilayered printed wiring boards .100 (2.54 mm) spacing.

Ⓐ denotes changes

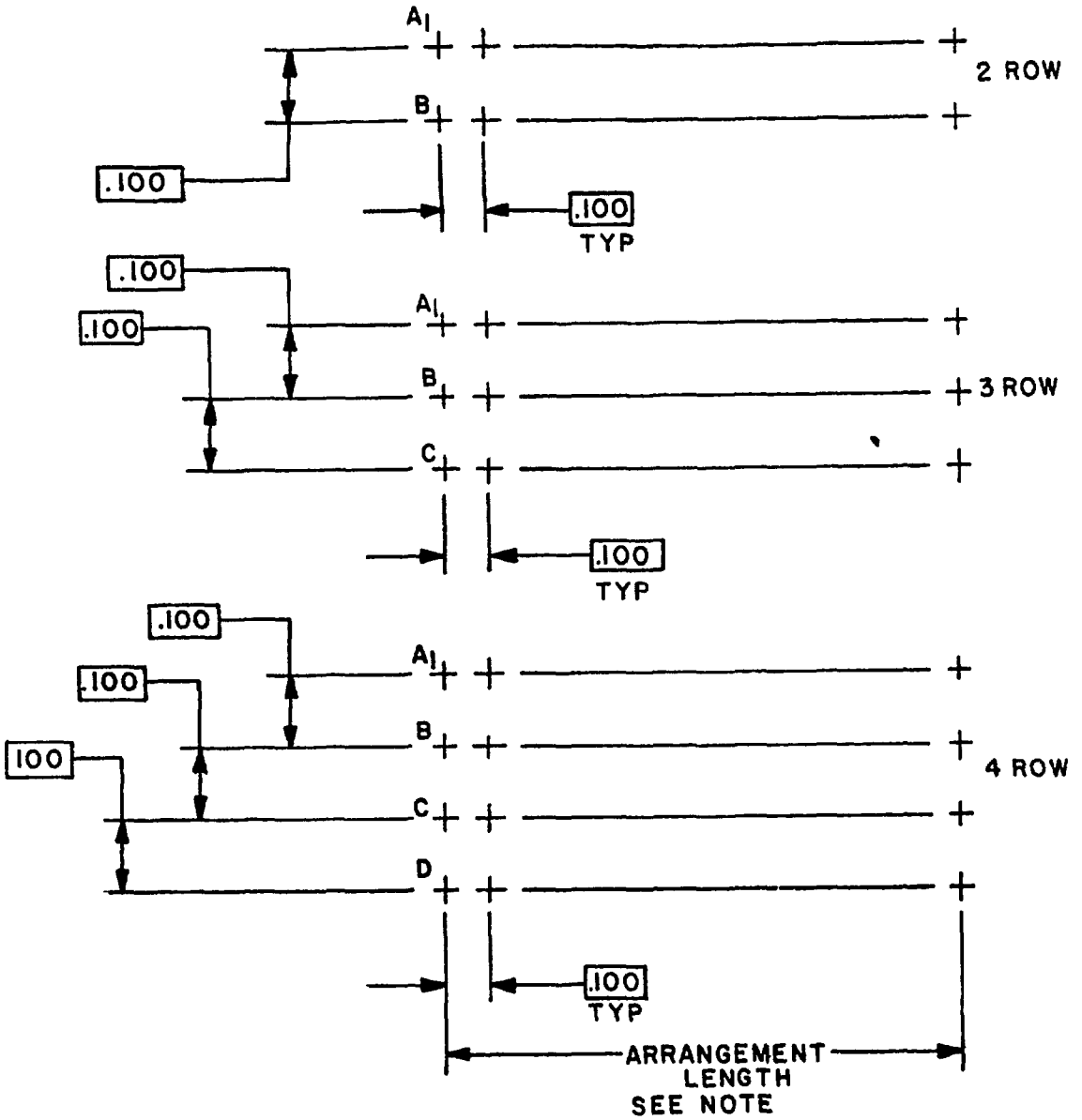
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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.075	1.91	.490	12.45
.003	0.08	.112	2.84	.540	13.72
.004	0.10	.220	5.59	.590	14.99
.005	0.13	.235	5.97	.650	16.51
.010	0.25	.300	7.62	.670	17.02
.021	0.53	.335	8.51	.785	19.94
.040	1.02	.390	9.91		
.060	1.52	.435	11.05		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. These connectors mate with connectors specified in MIL-C-55302/166, MIL-C-55302/167 and MIL-C-55302/168 and are primarily for use with single-sided, double-sided, or multilayered printed wiring boards.
5. The polarization feature, utilizing four M55302/78-02 polarizing keys which are not supplied with the connector, will provide 256 unique polarizing positions.
6. For dimensions A through K, see table I.
7. Crimp contacts only are removable and replaceable from the rear.
8. When crimp contacts are specified (indicators 1 and 2, table II), a quantity of at least 102 percent of the normal complement but not less than 2 spare contacts shall be supplied unassembled with each connector. When PCB contacts are specified (indicator 4, Table II), they will be supplied assembled into each connector.
9. Locking screws are not supplied with the connector. See MIL-C-55302/172.
10. Contact rows are identified by molded letters on the rear face. Numbers indicating every five cavities per row appear on rear face except for arrangements consisting of 10 contacts per row which lack numerals. Up to 9 contact locations at the high numeral end of the contact row will not be identified.
11. Indicated dimension shall occur at least at one point within each of the following zones. The zones are defined as areas of the feature containing contact positions 1 through 5, 16 through 20, 46 through 50, 76 through 80, and the zone containing the highest five contact positions. The tolerance for the areas not described above shall be +.010 (0.25 mm) -.004 (0.10 mm).
12. Dimension pertains when contact is seated against retention system rearward.
13. Perfect form at MMC not required. Dimension for user installation purposes only.

FIGURE 1. Connector, female brush, straight-through for multilayered printed wiring boards .100 (2.54 mm) spacing - Continued.

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NOTES:
1. The arrangement length is determined by subtracting one contact from the number of contacts per row (see table I) and multiplying the result by 0.100 inch (2.54 mm).
2. Arrangement length = 0.100 (2.54 mm) times (N-1), where N equals the number of contacts per row.

FIGURE 2. Insert arrangement for .100 (2.54 mm) contact spacing, female brush connectors (engaging face).

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Crimp contacts: All crimp contacts shall be in accordance with MIL-C-55302/171.

PCB contacts: All PCB contacts shall contain 7 brush wires with a tolerance limit of ±1 providing the average outgoing quality limit (AOQL) at each tolerance extreme does not exceed 1 percent. Each brush wire diameter shall be .0073 to .0085 (0.185 mm to 0.216 mm).

Material:

Body: Glass filled thermoplastic molding material in accordance with MIL-M-24519, type GPT-15F.

Crimp contacts: In accordance with MIL-C-55302/171.

PCB contacts: In accordance with MIL-C-55302.

Plating: The contact termination plating shall be in accordance with table II. Brush wire plating shall be gold in accordance with MIL-G-45204, type II, grade C, class 1 over nickel in accordance with QQ-N-290. Cut ends of brush wires need not be plated.

Contact identification: See figure 1 (note 10) and figure 2.

Oversized pin exclusion: Not applicable.

Contact engagement and separation forces: Not applicable, see mating and unmating.

Mating and unmating: The maximum mating and unmating force shall be 1.5 ounces times the total number of contacts.

Ⓐ Contact current rating: The connector may have any combination of current flow and ambient temperature provided the contact or connector temperature does not exceed 125°C. The test current shall be 5 amperes for individually connected contacts and 3 amperes for series wired contacts.

Contact resistance: The average resistance of all contact pairs measured on row A contacts shall not exceed 0.016 ohm, and no individual pair shall have a resistance exceeding 0.020 ohm. To allow for increased terminal lengths, add 0.001 ohm for each contact row beyond row A. The point of measurement shall be the point of terminal exit from the printed wiring board.

Contact retention: 7 pounds per contact minimum when the force is applied in the rearward direction. The displacement shall not exceed 0.008 (0.20 mm). The displacement shall be recorded after a 1 pound preload is used to determine the zero displacement reference.

Dielectric withstanding voltage:

Sea level: 1,300 volts rms, 60 Hz, ac.

High altitude: 325 volts rms, 60 Hz, ac.

Extended contact life: Connector shall meet the performance requirements of MIL-C-55302, contact life, when subjected to 20,000 insertion and withdrawal cycles during qualification inspection. Locking screws shall not be attached during contact life cycling.

Insert arrangement: See figure 2.

Contact replacements (crimp only): See MIL-C-55302/171.

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Contact tools for crimp only:

Extraction tool: See MS27509R22.

Insertion tool: See MS27509A22.

Part number: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled as follows:

Example:

M55302/169 B 18 2

Specification sheet number

Number of rows (indicator)
(see table I for row indicator)

Number of contacts per row
(see table I for contact indicator)

Gold plated contact tails
(see table II for indicator)

User instructions: Each shipment of connectors shall include, as a minimum, instructions for: Locking screw usage, polarization key usage and electrical probing information.

TABLE I. Indicators and dimensions. Ⓐ

2 Contact rows		3 Contact rows		4 Contact rows		Dimensions 1/		
Indicator	Total contacts	Indicator	Total contacts	Indicator	Total contacts	A Max	|B|	+.010 K -.000
A10 2/	20	B10 2/	30	C10 2/	40	2.295	1.975	1.120
A11	22	B11	33	C11	44	2.395	2.075	1.220
A12	24	B12	36	C12	48	2.495	2.175	1.320
A13	26	B13	39	C13	52	2.595	2.275	1.420
A14	28	B14	42	C14	56	2.695	2.375	1.520
A15	30	B15	45	C15	60	2.795	2.475	1.620
A16	32	B16	48	C16	64	2.895	2.575	1.720
A17	34	B17	51	C17	68	2.995	2.675	1.820
A18	36	B18	54	C18	72	3.095	2.775	1.920
A19	38	B19	57	C19	76	3.195	2.875	2.020
A20	40	B20	60	C20	80	3.295	2.975	2.120
A21	42	B21	63	C21	84	3.395	3.075	2.220
A22	44	B22	66	C22	88	3.495	3.175	2.320
A23	46	B23	69	C23	92	3.595	3.275	2.420
A24	48	B24	72	C24	96	3.695	3.375	2.520
A25	50	B25	75	C25	100	3.795	3.475	2.620
A26	52	B26	78	C26	104	3.895	3.575	2.720
A27	54	B27	81	C27	108	3.995	3.675	2.820
A28	56	B28	84	C28	112	4.095	3.775	2.920
A29	58	B29	87	C29	116	4.195	3.875	3.020
A30	60	B30	90	C30	120	4.295	3.975	3.120
A31	62	B31	93	C31	124	4.395	4.075	3.220
A32	64	B32	96	C32	128	4.495	4.175	3.320
A33	66	B33	99	C33	132	4.595	4.275	3.420
A34	68	B34	102	C34	136	4.695	4.375	3.520
A35	70	B35	105	C35	140	4.795	4.475	3.620
A36	72	B36	108	C36	144	4.895	4.575	3.720

See footnotes at end of table.

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TABLE I. Indicators and dimensions - Continued. Ⓐ

2 Contact rows		3 Contact rows		4 Contact rows		Dimensions 1/		
Indicator 2/	Total contacts	Indicator 2/	Total contacts	Indicator 2/	Total contacts	A Max	|B|	+.010 K -.000
A37	74	B37	111	C37	148	4.995	4.675	3.820
A38	76	B38	114	C38	152	5.095	4.775	3.920
A39	78	B39	117	C39	156	5.195	4.875	4.020
A40	80	B40	120	C40	160	5.295	4.975	4.120
A41	82	B41	123	C41	164	5.395	5.075	4.220
A42	84	B42	126	C42	168	5.495	5.175	4.320
A43	86	B43	129	C43	172	5.595	5.275	4.420
A44	88	B44	132	C44	176	5.695	5.375	4.520
A45	90	B45	135	C45	180	5.795	5.475	4.620
A46	92	B46	138	C46	184	5.895	5.575	4.720
A47	94	B47	141	C47	188	5.995	5.675	4.820
A48	96	B48	144	C48	192	6.095	5.775	4.920
A49	98	B49	147	C49	196	6.195	5.875	5.020
A50	100	B50	150	C50	200	6.295	5.975	5.120
A51	102	B51	153	C51	204	6.395	6.075	5.220
A52	104	B52	156	C52	208	6.495	6.175	5.320
A53	106	B53	159	C53	212	6.595	6.275	5.420
A54	108	B54	162	C54	216	6.695	6.375	5.520
A55	110	B55	165	C55	220	6.795	6.475	5.620
A56	112	B56	168	C56	224	6.895	6.575	5.720
A57	114	B57	171	C57	228	6.995	6.675	5.820
A58	116	B58	174	C58	232	7.095	6.775	5.920
A59	118	B59	177	C59	236	7.195	6.875	6.020
A60	120	B60	180	C60	240	7.295	6.975	6.120
A61	122	B61	183	C61	244	7.395	7.075	6.220
A62	124	B62	186	C62	248	7.495	7.175	6.320
A63	126	B63	189	C63	252	7.595	7.275	6.420
A64	128	B64	192	C64	256	7.695	7.375	6.520
A65	130	B65	195	C65	260	7.795	7.475	6.620
A66	132	B66	198	C66	264	7.895	7.575	6.720
A67	134	B67	201	C67	268	7.995	7.675	6.820
A68	136	B68	204	C68	272	8.095	7.775	6.920
A69	138	B69	207	C69	276	8.195	7.875	7.020
A70	140	B70	210	C70	280	8.295	7.975	7.120
A71	142	B71	213	C71	284	8.395	8.075	7.220
A72	144	B72	216	C72	288	8.495	8.175	7.320
A73	146	B73	219	C73	292	8.595	8.275	7.420
A74	148	B74	222	C74	296	8.695	8.375	7.520
A75	150	B75	225	C75	300	8.795	8.475	7.620
A76	152	B76	228	C76	304	8.895	8.575	7.720
A77	154	B77	231	C77	308	8.995	8.675	7.820
A78	156	B78	234	C78	312	9.095	8.775	7.920
A79	158	B79	237	C79	316	9.195	8.875	8.020
A80	160	B80	240	C80	320	9.295	8.975	8.120
A81	162	B81	243	C81	324	9.395	9.075	8.220
A82	164	B82	246	C82	328	9.495	9.175	8.320
A83	166	B83	249	C83	332	9.595	9.275	8.420
A84	168	B84	252	C84	336	9.695	9.375	8.520
A85	170	B85	255	C85	340	9.795	9.475	8.620
A86	172	B86	258	C86	344	9.895	9.575	8.720
A87	174	B87	261	C87	348	9.995	9.675	8.820
A88	176	B88	264	C88	352	10.095	9.775	8.920
A89	178	B89	267	C89	356	10.195	9.875	9.020
A90	180	B90	270	C90	360	10.295	9.975	9.120
A91	182	B91	273	C91	364	10.395	10.075	9.220
A92	184	B92	276	C92	368	10.495	10.175	9.320
A93	186	B93	279	C93	372	10.595	10.275	9.420

See footnotes at end of table.

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TABLE I. Indicators and dimensions - Continued. Ⓐ

2 Contact rows		3 Contact rows		4 Contact rows		Dimensions 1/		
Indicator 2/	Total contacts	Indicator 2/	Total contacts	Indicator 2/	Total contacts	A Max	|B|	+.010 K -.000
A94	188	B94	282	C94	376	10.695	10.375	9.520
A95	190	B95	285	C95	380	10.795	10.475	9.620
A96	192	B96	288	C96	384	10.895	10.575	9.720
A97	194	B97	291	C97	388	10.995	10.675	9.820
A98	196	B98	294	C98	392	11.095	10.775	9.920
A99	198	B99	297	C99	396	11.195	10.875	10.020
A00	200	B00	300	C00	400	11.295	10.975	10.120

1/ All dimensions are in inches. Metric equivalents may be calculated based on 1 inch = 25.4 mm.
2/ Indicators consist of, a letter designating number of contact rows and two digits indicating number of contacts per row. Indicators A00, B00, and C00 denote 100 contacts per row.

TABLE II. Contact option.

	Indicator	Option	Contact type
Ⓐ	1	Termination plating: Tin lead per MIL-P-81728, 50 to 70 percent tin, .0001 minimum thick over copper per MIL-C-14550	Crimp
Ⓐ	2	Termination plating: Gold per MIL-G-45204, type II, grade C, class 1 over copper per MIL-C-14550	Crimp
	3	Connectors supplied less contacts	-----
Ⓐ	4	Termination plating: Tin lead per MIL-P-81728, 50 to 70 percent tin, .0001 minimum thick over copper per MIL-C-14550	Printed circuit board w/.145 ±.025 stickout

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 85, 99

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

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DLA - ES

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